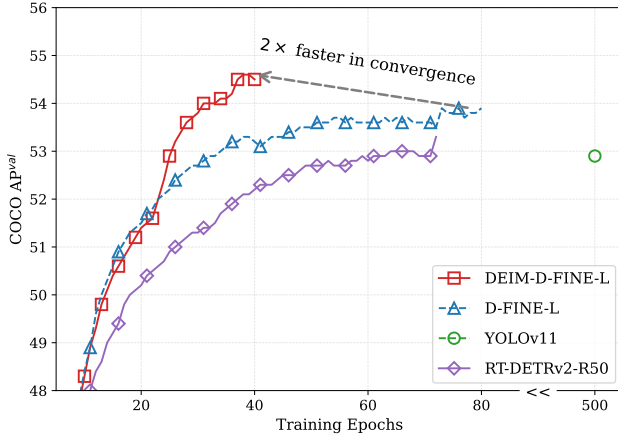


DEIM: DETR with Improved Matching for Fast Convergence

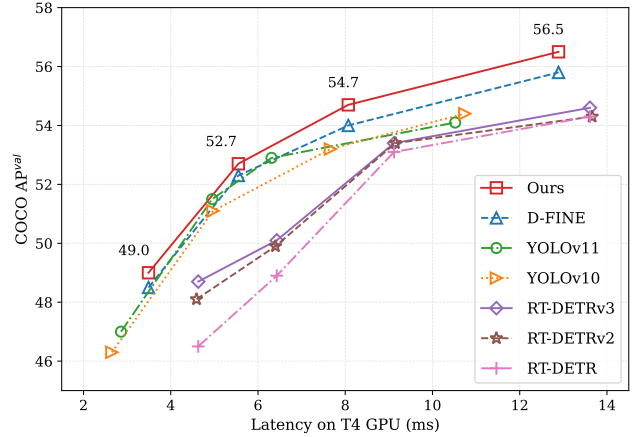
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(a) **Faster**: training is more compute-efficient



(b) **Better**: exceeding all real-time detectors

Figure 1. Comparison with state-of-the-art real-time object detectors on COCO [20]. The proposed DEIM achieves faster convergence (a) and superior performance in terms of average precision (AP) and latency (b) when compared to state-of-the-art real-time object detectors.

Abstract

We introduce *DEIM*, an innovative and efficient training framework designed to accelerate convergence in real-time object detection with Transformer-based architectures (DETR). To mitigate the sparse supervision inherent in one-to-one (O2O) matching in DETR models, *DEIM* employs a Dense O2O matching strategy. This approach increases the number of positive samples per image by incorporating additional targets, using standard data augmentation techniques. While Dense O2O matching speeds up convergence, it also introduces numerous low-quality matches that could affect performance. To address this, we propose the *Matchability-Aware Loss (MAL)*, a novel loss function that optimizes matches across various quality levels, enhancing the effectiveness of Dense O2O. Extensive experiments on the COCO dataset validate the efficacy of *DEIM*. When integrated with RT-DETR and D-FINE, it consistently boosts performance while reducing training time by 50%. Notably, paired with RT-DETRv2, *DEIM* achieves 53.2% AP in a single day of training on an NVIDIA 4090 GPU. Additionally, *DEIM*-trained real-time models outperform leading real-time object detectors, with *DEIM-D-FINE-L* and

DEIM-D-FINE-X achieving 54.7% and 56.5% AP at 124 and 78 FPS on an NVIDIA T4 GPU, respectively, without the need for additional data. We believe *DEIM* sets a new baseline for advancements in real-time object detection. Our code and pre-trained models are available at <https://www.shihuahuang.cn/DEIM/>.

1. Introduction

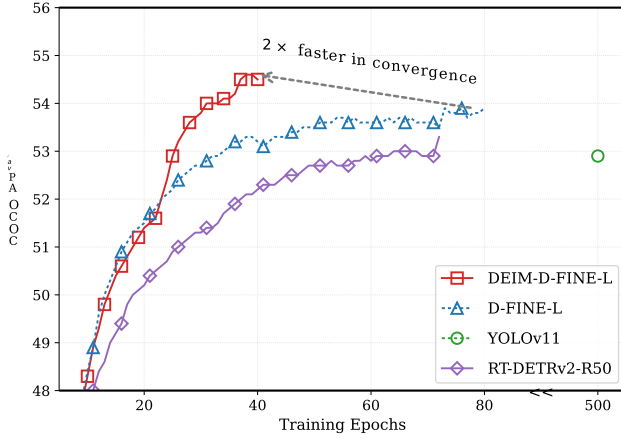
Object detection is a fundamental task in computer vision, widely applied in fields like autonomous driving [5, 6], robot navigation [9], etc. The growing demand for efficient detectors has spurred the development of real-time detection methods. In particular, YOLO emerges as one of the main paradigms for real-time object detection, owing to its compelling trade-off between latency and accuracy [1, 28, 32, 34, 44]. YOLO models are widely recognized as one-stage detectors based on convolutional neural networks. *One-to-many (O2M)* assignment strategy has been widely used in YOLO series [1, 28, 34, 44], where each target box is associated with multiple anchors. This strategy is known to be effective, as it provides dense supervision signals, which accelerate convergence and enhance

DEIM: 具有改进匹配机制的DETR, 实现快速收敛

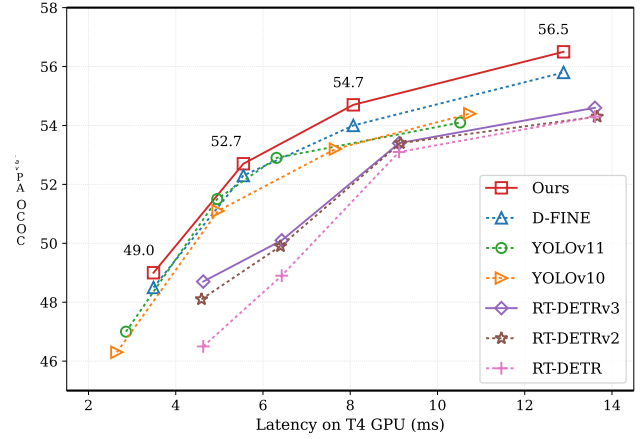
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(a) **Faster**: training is more compute-efficient



(b) **Better**: exceeding all real-time detectors

图1. 与COCO数据集[20]上最先进的实时目标检测器对比。所提出的DEIM方法在收敛速度 (a) 以及平均精度 (AP) 与延迟 (b) 方面均表现出更快的收敛性和更优的性能, 优于当前最先进的实时目标检测器。

摘要

We introduce DEIM, an innovative and efficient training framework designed to accelerate convergence in real-time object detection with Transformer-based architectures (DETR). To mitigate the sparse supervision inherent in one-to-one (O2O) matching in DETR models, DEIM employs a Dense O2O matching strategy. This approach increases the number of positive samples per image by incorporating additional targets, using standard data augmentation techniques. While Dense O2O matching speeds up convergence, it also introduces numerous low-quality matches that could affect performance. To address this, we propose the Matchability-Aware Loss (MAL), a novel loss function that optimizes matches across various quality levels, enhancing the effectiveness of Dense O2O. Extensive experiments on the COCO dataset validate the efficacy of DEIM. When integrated with RT-DETR and D-FINE, it consistently boosts performance while reducing training time by 50%. Notably, paired with RT-DETRv2, DEIM achieves 53.2% AP in a single day of training on an NVIDIA 4090 GPU. Additionally, DEIM-trained real-time models outperform leading real-time object detectors, with DEIM-D-FINE-L and

DEIM-D-FINE-X achieving 54.7% and 56.5% AP at 124 and 78 FPS on an NVIDIA T4 GPU, respectively, without the need for additional data. We believe DEIM sets a new baseline for advancements in real-time object detection. Our code and pre-trained models are available at <https://www.shihuahuang.cn/DEIM/>.

1. 引言

目标检测是计算机视觉中的一项基础任务, 广泛应用于自动驾驶[5, 6]、机器人导航[9]等领域。对高效检测器日益增长的需求推动了实时检测方法的发展。其中, YOLO因其其在延迟与精度之间出色的平衡[1, 28, 32, 34, 44], 成为实时目标检测的主要范式之一。YOLO模型被广泛认为是基于卷积神经网络的一阶段检测器。One-to-many (O2M)分配策略在YOLO系列[1, 28, 34, 44]中被广泛采用, 该策略将每个目标框与多个锚框关联。这种策略被证实是有效的, 因为它提供了密集的监督信号, 从而加速收敛并提升性能。

performance [44]. However, it produces multiple overlapping bounding boxes per object, requiring a hand-crafted Non-Maximum Suppression (NMS) to remove redundancies, introducing latency and instability [32, 43].

The advent of Transformer-based detection (DETR) paradigm [3] has attracted significant attention [4, 39, 46], leveraging multi-head attention to capture global context, thereby enhancing localization and classification. DETRs adopt a *one-to-one* (O2O) matching strategy that leverages the Hungarian [16] algorithm to establish a unique correspondence between predicted boxes and the ground-truth objects during training, eliminating the need for NMS. This end-to-end framework offers a compelling alternative for real-time object detection.

However, **slow convergence** remains one of the primary limitations of DETRs, and we hypothesize that the reasons are two-fold. ❶ *Sparse supervision*: The O2O matching mechanism assigns only one positive sample per target, greatly limiting the number of positive samples. In contrast, O2M generates several times more positive samples. This scarcity of positive samples restricts dense supervision, which impedes effective model learning—particularly for small objects, where dense supervision is crucial for performance. ❷ *Low-quality matches*: Unlike traditional methods that rely on dense anchors (usually > 8000), DETR employs a small number (100 or 300) of randomly initialized queries. These queries lack spatial alignment with targets, leading to numerous low-quality matches in the training, where matched boxes have low IoU with the targets but high confidence scores.

To address the scarcity of supervision in DETR, recent studies have relaxed the constraints of O2O matching by incorporating O2M assignments into O2O training, thereby introducing auxiliary positive samples per target to increase supervision. Group DETR [4] achieves this by using multiple query groups, each with independent O2O matching, while Co-DETR [46] incorporates O2M methods from object detectors like Faster R-CNN [29] and FCOS [31]. Although these approaches successfully increase the number of positive samples, they also require additional decoders, which increases computational overhead and risks generating redundant high-quality predictions as traditional detectors. In contrast, we propose a novel yet straightforward approach named dense one-to-one (Dense O2O) matching. Our key idea is to increase the number of targets in each training image, which in turn generates more positive samples during the training. Notably, this can be easily achieved using classical techniques such as mosaic [1] and mixup [38] augmentations, which generates additional positive samples per image while preserving the one-to-one matching framework. Dense O2O matching can provide a level of supervision comparable to O2M approaches, without the added complexity and overhead typically associated

with O2M methods.

Despite attempts to improve query initialization using priors [18, 39, 43, 45], which enable more effective query distributions around objects. These improved initialization methods, often relying on limited feature information extracted from the encoder [39, 43], tend to cluster queries around a few prominent objects. In contrast, most non-salient objects lack nearby queries, leading to low-quality matches. This issue becomes even more pronounced when using Dense O2O. As the number of targets increases, the disparity between prominent and non-prominent targets grows, leading to a rise in low-quality matches despite the overall increase in matching quantity. In this case, if the loss function has limitations in handling these low-quality matches, this disparity will persist, hindering the model from achieving better performance.

Existing loss functions [19, 40] in DETRs, such as Vari-focal Loss (VFL) [40], are tailored to dense anchors where the number of low-quality matches is relatively low. They primarily penalize high-quality matches, especially matches with high IoU but low confidence, and discard low-quality matches. To address low-quality matches and further improve Dense O2O, we propose Matchability-Aware Loss (MAL). MAL scales the penalty based on matchability by incorporating the IoU between matched queries and targets with classification confidence. MAL performs similarly to VFL for high-quality matches but places greater emphasis on low-quality matches, improving the utility of limited positive samples during training. Furthermore, MAL provides a simpler mathematical formulation than VFL.

The proposed DEIM combines Dense O2O with MAL to create an effective training framework. We conducted extensive experiments on the COCO [20] dataset to evaluate the effectiveness of DEIM. The results in Fig. 1 (a) show that DEIM significantly accelerates the convergence of RT-DETRv2 [24] and D-FINE [27] and achieves improved performance as well. Specifically, with only half the number of training epochs, our method outperforms RT-DETRv2 and D-FINE by 0.2 and 0.6 AP, respectively. Additionally, our approach enables training a ResNet50-based DETR model on a single 4090 GPU, achieving 53.2% mAP within a single day (approximately 24 epochs). By incorporating more efficient models, we also introduce a new set of real-time detectors that outperform existing models, including the latest YOLOv11 [13], setting a new state-of-the-art (SoTA) for real-time object detection (Fig. 1 (b)).

The main contributions of this work are summarized as follows:

- We introduce DEIM, a simple and flexible training framework for real-time object detection.
- DEIM accelerates the convergence by improving the quantity and quality of matching with Dense O2O and MAL, respectively.

性能[44]。然而，它会对每个物体产生多个重叠的边界框，需要手动设计非极大值抑制（NMS）来消除冗余，这引入了延迟和不稳定性[32, 43]。

基于Transformer的检测（DETR）范式[3]的出现引起了广泛关注[4, 39, 46]，它利用多头注意力机制捕捉全局上下文，从而提升定位与分类性能。DETR采用一种*one-to-one (O2O)*匹配策略，在训练过程中借助匈牙利算法[16]建立预测框与真实目标之间的一一对应关系，无需非极大值抑制。这种端到端框架为实时目标检测提供了一个极具吸引力的替代方案。

然而，收敛速度慢仍然是DETR的主要限制之一，我们推测其原因有二。① *Sparse supervision*：一对一匹配机制每个目标仅分配一个正样本，极大限制了正样本的数量。相比之下，一对多匹配能产生数倍于前者的正样本。正样本的稀缺制约了密集监督，从而阻碍了模型的有效学习——尤其对于小目标而言，密集监督对性能至关重要。② *Low-quality matches*：不同于依赖密集锚点（通常为> 8000个）的传统方法，DETR采用少量（100或300个）随机初始化的查询向量。这些查询与目标之间缺乏空间对齐，导致训练中出现大量低质量匹配——被匹配的边界框与目标交并比较低，却具有高置信度得分。

为了解决DETR中监督信号稀缺的问题，近期研究通过将一对多（O2M）分配融入一对一（O2O）训练，放宽了O2O匹配的限制，从而为每个目标引入辅助正样本以增强监督。Group DETR [4]通过使用多个查询组实现这一目标，每个组进行独立的O2O匹配；而Co-DETR [46]则融合了来自Faster R-CNN [29]和FCOS [31]等目标检测器中的O2M方法。尽管这些方法成功增加了正样本数量，但它们也需要额外的解码器，这会增加计算开销，并可能像传统检测器那样产生冗余的高质量预测。相比之下，我们提出了一种新颖而直接的方法——密集一对一（Dense O2O）匹配。我们的核心思想是增加每张训练图像中的目标数量，进而在训练过程中生成更多正样本。值得注意的是，这可以通过马赛克增强[1]和混合增强[38]等经典技术轻松实现，这些方法能在保持一对一匹配框架的同时，为每张图像生成额外的正样本。Dense O2O匹配能够提供与O2M方法相媲美的监督水平，同时避免了通常伴随而来的复杂性和额外开销。

使用O2M方法。

尽管已有尝试通过先验知识改进查询初始化[18, 39, 43, 45]，使查询能更有效地分布在物体周围，但这些改进方法通常依赖从编码器提取的有限特征信息[39, 43]，容易使查询聚集在少数显著物体周围。相比之下，大多数非显著物体因附近缺乏查询而难以获得高质量匹配。在使用密集一对一匹配时，这一问题尤为突出：随着目标数量增加，显著与非显著目标间的差距扩大，导致低质量匹配数量随匹配总量上升而增长。此时若损失函数处理低质量匹配的能力有限，这种差距将持续存在，阻碍模型获得更优性能。

现有DETR中的损失函数[19, 40]，如Varifocal Loss (VFL) [40]，是为密集锚点设计的，其中低质量匹配的数量相对较少。它们主要惩罚高质量匹配，特别是具有高IoU但低置信度的匹配，并丢弃低质量匹配。为了解决低质量匹配并进一步改进Dense O2O，我们提出了匹配能力感知损失（MAL）。MAL通过结合匹配查询与目标之间的IoU以及分类置信度，根据匹配能力来调整惩罚力度。对于高质量匹配，MAL的表现与VFL类似，但更强调低质量匹配，从而提高了训练过程中有限正样本的利用率。此外，MAL提供了比VFL更简洁的数学公式。

提出的DEIM将Dense O2O与MAL相结合，构建了一个高效的训练框架。我们在COCO [20]数据集上进行了大量实验以评估DEIM的有效性。图1(a)中的结果表明，DEIM显著加速了RT-DETRv2 [24]和D-FINE [27]的收敛，并实现了性能提升。具体而言，仅使用一半的训练周期数，我们的方法在AP指标上分别超过RT-DETRv2和D-FINE 0.2和0.6。此外，我们的方法使得基于ResNet50的DETR模型能够在单张4090 GPU上进行训练，在一天内（约24个周期）达到53.2%的mAP。通过引入更高效的模型，我们还提出了一系列新的实时检测器，其性能超越现有模型（包括最新的YOLOv11 [13]），为实时目标检测树立了新的最先进水平（SoTA）（图1(b)）。

本工作的主要贡献总结如下：

- 我们介绍DEIM，一个简单灵活的训练框架，用于实时目标检测。
- DEIM通过分别利用Dense O2O和MAL提升匹配的数量与质量，从而加速收敛过程。

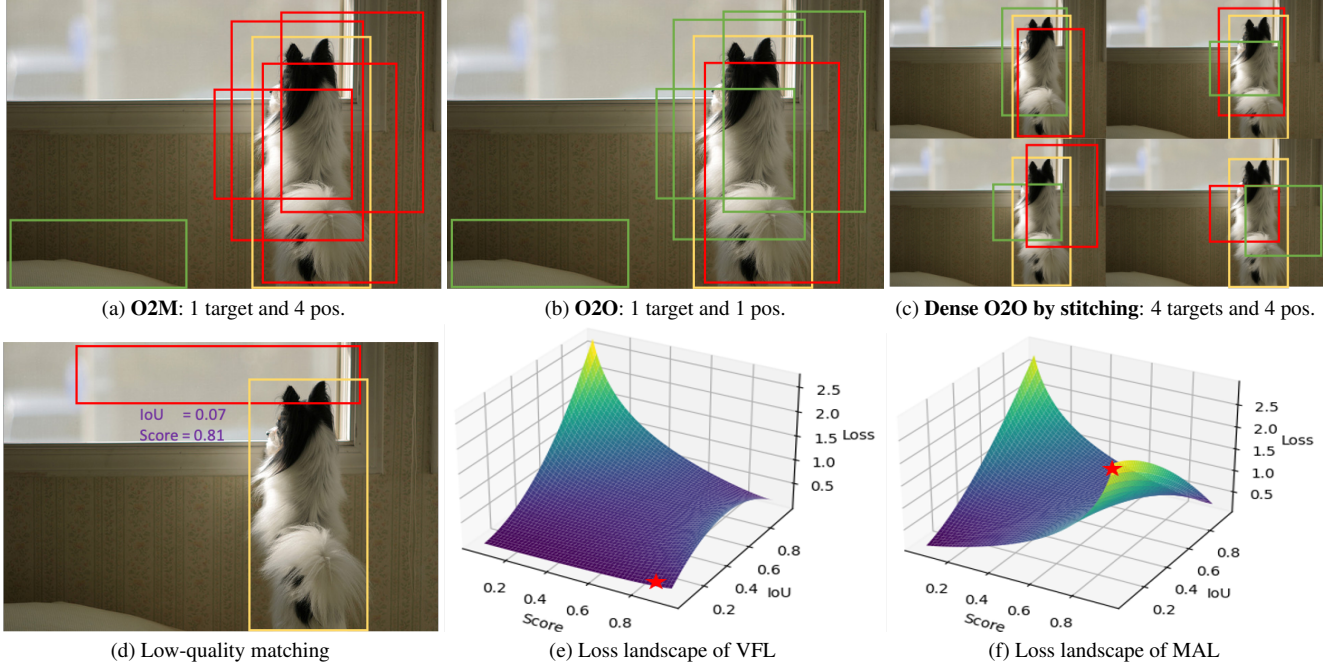


Figure 2. **An illustration of our proposed DEIM.** Yellow, red, and green boxes represent the GT, positive and negative samples, respectively. ‘pos.’ denotes the positive samples. *Top:* Our Dense O2O (Fig. 2c) can provide the same quality of positive samples as O2M (Fig. 2a). *Bottom:* For the low-quality matching, its loss values when using VFL [40] and MAL are marked by $\star$, indicating MAL can optimize those cases more effectively.

- With our method, existing real-time DETRs achieve better performance while halving training costs. Specifically, our method exceeds YOLOs and establishes a new SoTA in real-time object detection after being paired with efficient models in D-FINE.

2. Related Work

Object detection with transformer (DETR) [3] represents a shift from traditional CNN architectures to transformers. By using Hungarian [16] algorithm for one-to-one matching, DETR eliminates the need for hand-crafted NMS as the post-processing and enables end-to-end object detection. However, it suffers from slow convergence and dense computation.

Increasing positive samples. One-to-one matching limits each target to a single positive sample, providing far less supervision than O2M and hindering the optimization. Some studies have explored ways to increase supervision within the O2O framework. Group DETR [4], for instance, employs the concept of “groups” to approximate the O2M. It uses K groups of queries, where $K > 1$, and performs O2O matching independently within each group. This allows each target to be assigned K positive samples. However, to prevent communication between groups, each group requires a separate decoder layer, ultimately resulting in K parallel decoders. The hybrid matching

scheme in H-DETR [15] works similarly to Group DETR. Co-DETR [46] reveals that a one-to-many assignment approach helps the model learn more distinctive feature information, so it proposed a collaborative hybrid assignment scheme to enhance encoder representations through auxiliary heads with one-to-many label assignments, like Faster R-CNN [29] and FCOS [31]. The existing methods aim to increase the number of positive samples per target to enhance supervision. In contrast, Our Dense O2O explores another direction — increasing the number of targets per training image to boost supervision effectively. Unlike existing methods, which require additional decoders or heads and thus increase training resource consumption, our approach is computation-free.

Optimizing low-quality matches. The sparse and randomly initialized queries lack spatial alignment with targets, resulting in a high proportion of low-quality matches that impede model convergence. Several methods have introduced prior knowledge into query initialization, such as anchor queries [35], DAB-DETR [21], DN-DETR [18], and dense distinct queries [41]. More recently, inspired by two-stage paradigms [29, 45], methods like DINO [39] and RT-DETR [43] leverage top-ranked predictions from the encoder’s dense outputs to refine decoder queries [36]. These strategies enable more effective query initialization closer to target regions. However, low-quality matches persist as

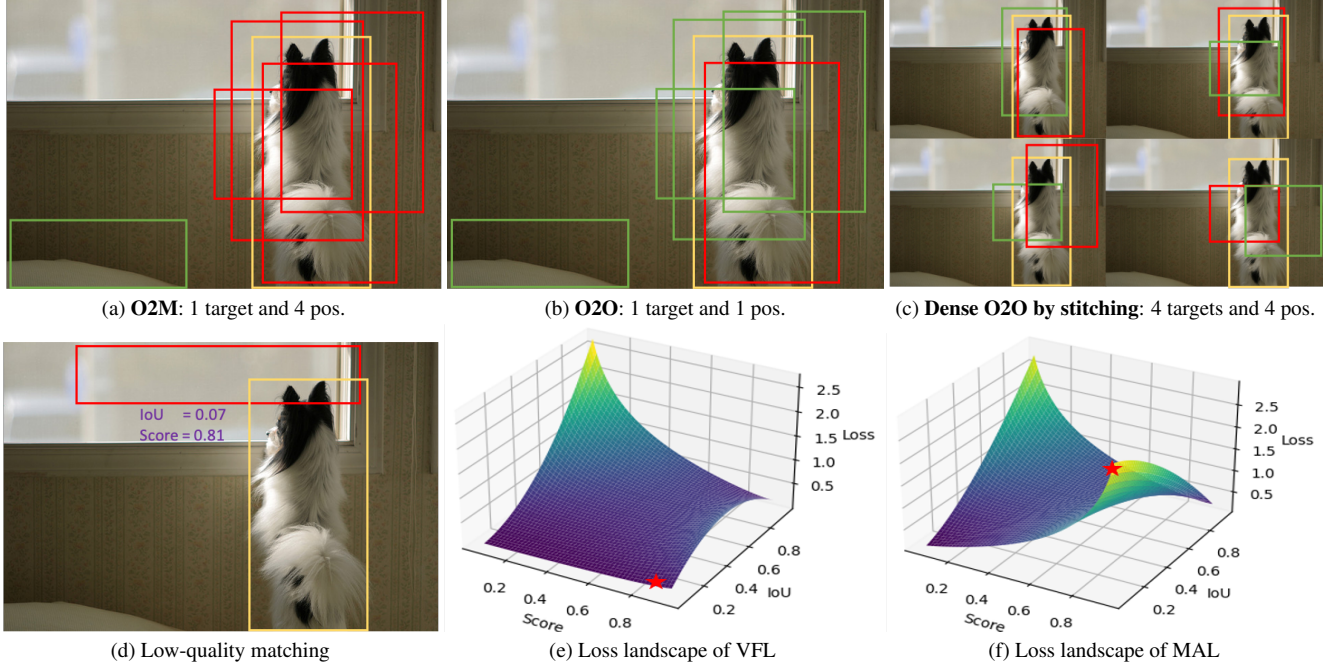


图2. 我们提出的DEIM示意图。黄色、红色和绿色框分别代表GT、正样本和负样本。"pos."表示正样本。*Top*: 我们的密集O2O（图2c）能提供与O2M（图2a）同等质量的正样本。*Bottom*: 对于低质量匹配，使用VFL[40]和MAL时的损失值分别标记为★，表明MAL能更有效地优化这些情况。

- 通过我们的方法，现有的实时DETR模型在训练成本减半的同时实现了更优性能。具体而言，该方法在与D-FINE中的高效模型结合后，性能超越了YOLO系列，并在实时目标检测领域建立了新的技术标杆{v*}。

2. 相关工作

基于Transformer的目标检测（DETR）[3]代表了从传统CNN架构向Transformer的转变。通过使用匈牙利算法[16]进行一对一匹配，DETR消除了后处理中对人工设计非极大值抑制的需求，实现了端到端的目标检测。然而，该方法存在收敛速度慢和计算密集的问题。

增加正样本。一对一匹配将每个目标限制为单个正样本，提供的监督信息远少于一对多匹配，从而阻碍了优化。一些研究探索了在一对一框架内增强监督的方法。例如，Group DETR [4] 采用“分组”概念来近似实现一对多匹配。它使用 K 组查询（其中 $K > 1$ ），并在每个组内独立执行一对一匹配。这使得每个目标可以被分配 K 个正样本。然而，为了防止组间通信，每个组都需要独立的解码器层，最终形成了 K 个并行解码器。混合匹配

H-DETR [15]中的方案与Group DETR的工作原理类似。Co-DETR [46]揭示了一对多分配方法有助于模型学习更具区分性的特征信息，因此它提出了一种协作混合分配方案，通过具有一对多标签分配的辅助头（如Faster R-CNN [29]和FCOS [31]）来增强编码器表示。现有方法旨在增加每个目标的正样本数量以增强监督。相比之下，我们的Dense O2O探索了另一个方向——增加每张训练图像中的目标数量，以有效提升监督效果。与现有方法需要额外解码器或头部从而增加训练资源消耗不同，我们的方法是无需计算开销的。

优化低质量匹配。稀疏且随机初始化的查询与目标之间缺乏空间对齐，导致低质量匹配比例较高，阻碍模型收敛。已有多种方法将先验知识引入查询初始化过程，例如锚点查询[35]、DAB-DETR[21]、DN-DETR[18]以及密集差异化查询[41]。近期，受两阶段范式[29,45]启发，DINO[39]和RT-DETR[43]等方法利用编码器密集输出中排名靠前的预测结果来优化解码器查询[36]。这些策略实现了更接近目标区域的有效查询初始化。然而，低质量匹配问题依然存在，因为{v*}

a significant challenge [22]. In RT-DETR [43], Varifocal Loss (VFL) is employed to reduce the uncertainty between classification confidence and box quality, enhancing real-time performance. Yet, VFL is primarily designed for traditional detectors with fewer low-quality matches and focuses on high-IoU optimization, leaving low-IoU matches under-optimized due to their minimal and flat loss values. Building on those advanced initializations, we introduce a matchability-aware loss to better optimize matches across varying quality levels, significantly enhancing the effectiveness of Dense O2O matching.

Reducing computation cost. Standard attention mechanisms involve dense computation. To improve efficiency and facilitate interactions with multi-scale features, several advanced attentions have been developed, such as deformable attention [45], multi-scale deformable attention [42], dynamic attention [7], and cascade window attention [37]. Additionally, recent research has focused on creating more efficient encoders. For example, Lite DETR [17] introduces an encoder block that interleaves updates between high-level and low-level features, while RT-DETR [43] combines CNN and self-attention in its encoder. Both designs significantly reduce resource consumption, especially RT-DETR. RT-DETR is the first real-time object detection model within the DETR framework. Building on this hybrid encoder, D-FINE [27] further optimizes RT-DETR with additional modules and refines the regression process by iteratively updating probability distributions instead of predicting fixed coordinates. This approach enables D-FINE to achieve a more favorable trade-off between latency and performance, slightly surpassing recent YOLO models. Leveraging these advancements in real-time DETRs, our method achieves impressive performance with reduced training costs, outperforming YOLO models by a substantial margin in real-time object detection.

3. Method

3.1. Preliminaries

O2M vs. O2O. The O2M assignment strategy [10, 44] is widely adopted in traditional object detectors, and its supervision can be formulated as follows:

$$\text{loss} = \sum_{i=0}^N \sum_{j=0}^{M_i} f(\hat{y}_{ij}, y_i), \quad (1)$$

where N is the total number of targets, M_i is the number of matches for the i -th target, $\hat{y}_{ij}$ represents the j -th match for the i -th target, y_i denotes the i -th ground-truth label, and f is the loss function. O2M enhances supervision by increasing M_i , i.e., assigning multiple queries to each target ($M_i > 1$) and thus providing dense supervision, as illustrated in Fig. 2a. In contrast, the O2O assignment only pairs

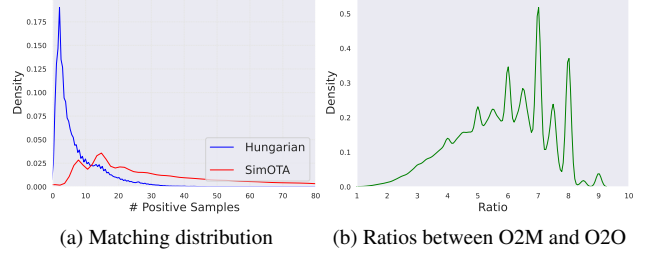


Figure 3. **Anchor/Query Match Comparison.** Comparison of the number of matched anchors/queries per image in one COCO epoch using one-to-many (SimOTA [44]) and one-to-one (Hungarian [3]) matching schemes.

each target with a single best prediction, determined via the Hungarian algorithm, which minimizes a cost function balancing classification and localization errors (Fig. 2b). O2O can be considered a special case of O2M where $M_i = 1$ for all targets.

Focal loss. Focal loss (FL) [19] was introduced to prevent an abundance of easy negatives from overwhelming the detector during training, directing focus instead towards a sparse set of hard examples. It serves as the default classification loss in DETRs [39, 45] and is defined as follows:

$$\text{FL}(p, y) = \begin{cases} -\alpha(1-p)^\gamma \log(p) & y = 1 \\ -(1-\alpha)p^\gamma \log(1-p) & y = 0, \end{cases} \quad (2)$$

where $y \in \{0, 1\}$ specifies the ground-truth class and $p \in [0, 1]$ represents the predicted probability for the foreground class. The parameter γ controls the balance between easy and hard samples, while α adjusts the weighting between foreground and background classes. In the FL, only the sample’s class and confidence are considered, with no attention given to bounding box quality, i.e., localization.

3.2. Improving matching efficiency: Dense O2O

The one-to-one (O2O) matching scheme, commonly used in DETR-based models, matches each target to only one predicted query. This approach, implemented via the Hungarian algorithm [16], allows for end-to-end training and eliminates the need for NMS. However, a key limitation of O2O is that it generates significantly fewer positive samples compared to traditional one-to-many (O2M) methods like SimOTA [44]. This leads to sparse supervision, which can slow down convergence during training.

To better understand this issue, we trained RT-DETRv2 [24] with a ResNet50 backbone on the MS COCO dataset [20]. We compared the number of positive matches generated by both Hungarian (O2O) and SimOTA (O2M) strategies. As shown in Fig. 3a, O2O produces a sharp peak under 10 positive matches per image, while O2M generates a broader distribution with many more positive matches,

一个重大挑战[22]。在RT-DETR[43]中，采用Varifocal Loss (VFL) 来降低分类置信度与框质量之间的不确定性，从而提升实时性能。然而，VFL主要针对传统检测器设计，这些检测器的低质量匹配较少，且VFL侧重于高IoU优化，导致低IoU匹配因损失值极小且平缓而优化不足。基于这些先进的初始化方法，我们引入了一种匹配感知损失，以更好地优化不同质量水平的匹配，显著提升了密集O2O匹配的有效性。

降低计算成本。标准注意力机制涉及密集计算。为提高效率并促进与多尺度特征的交互，已开发出多种先进注意力机制，如可变形注意力[45]、多尺度可变形注意力[42]、动态注意力[7]和级联窗口注意力[37]。此外，近期研究聚焦于构建更高效的编码器。例如，Lite DETR[17]引入了在高层与低层特征间交错更新的编码器模块，而RT-DETR[43]在其编码器中融合了CNN与自注意力机制。两种设计均显著降低了资源消耗，尤其是RT-DETR。RT-DETR是DETR框架内首个实时目标检测模型。基于这种混合编码器，D-FINE[27]通过附加模块进一步优化RT-DETR，并改进回归过程——通过迭代更新概率分布而非预测固定坐标。该方法使D-FINE在延迟与性能间取得更优平衡，小幅超越近期YOLO模型。借助实时DETR领域的这些进展，我们的方法以更低的训练成本实现了卓越性能，在实时目标检测任务中大幅超越YOLO模型。

3. 方法

3.1. 预备知识

O2M与O2O对比。O2M分配策略[10, 44]在传统目标检测器中被广泛采用，其监督可表述如下：

$$\text{loss} = \sum_{i=0}^N \sum_{j=0}^{M_i} f(\hat{y}_{ij}, y_i), \quad (1)$$

其中 N 是目标总数， M_i 是第 i 个目标的匹配数量， $\hat{y}_{ij}$ 代表第 i 个目标的第 j 次匹配， y_i 表示第 i 个真实标签， f 是损失函数。O2M通过增加 M_i 来增强监督，即为每个目标分配多个查询（ $M_i > 1$ ），从而提供密集监督，如图2a所示。相比之下，O2O分配仅将

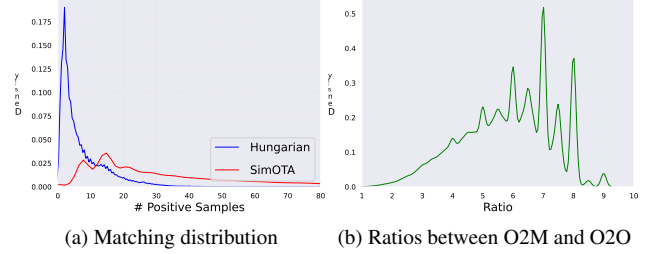


图3. 锚点/查询匹配对比。使用一对多（SimOTA [44]）和一对一（匈牙利算法 [3]）匹配方案，在COCO数据集的一个训练周期中每张图像匹配的锚点/查询数量对比。

每个目标都有一个单一的最佳预测，这是通过匈牙利算法确定的，该算法最小化了平衡分类和定位误差的成本函数（图2b）。O2O可以被视为O2M的一个特例，其中所有目标的 $\{v^*\}$ 均为1。

焦点损失。焦点损失（FL）[19]的提出是为了防止训练过程中大量易分类的负样本淹没检测器，从而将注意力转向稀疏的难例样本。它被用作 DETR [39, 45] 中的默认分类损失，其定义如下：

$$\text{FL}(p, y) = \begin{cases} -\alpha(1-p)^\gamma \log(p) & y = 1 \\ -(1-\alpha)p^\gamma \log(1-p) & y = 0, \end{cases} \quad (2)$$

其中 $y \in \{0, 1\}$ 指定了真实类别， $p \in [0, 1]$ 表示前景类的预测概率。参数 γ 控制简单样本与困难样本之间的平衡，而 α 则调整前景类与背景类之间的权重。在 FL 中，仅考虑样本的类别和置信度，未关注边界框质量，即定位准确性。

3.2. 提升匹配效率：密集O2O

一对一（O2O）匹配方案常用于基于DETR的模型中，它将每个目标仅与一个预测查询相匹配。这种方法通过匈牙利算法[16]实现，支持端到端训练，并无需非极大值抑制（NMS）。然而，O2O的一个关键局限在于，与SimOTA[44]等传统一对多（O2M）方法相比，其产生的正样本数量显著减少。这导致监督信号稀疏，可能延缓训练过程中的收敛速度。

为了更好地理解这个问题，我们在MS COCO数据集[20]上使用ResNet50骨干网络训练了RT-DETRv2[24]。我们比较了匈牙利（O2O）和SimOTA（O2M）两种策略产生的正匹配数量。如图3a所示，O2O在每张图像的正匹配数低于10时产生一个尖锐的峰值，而O2M则生成一个分布更广、正匹配数量显著更多的结果。

sometimes exceeding 80 positive samples for a single image. Fig. 3b further highlights that SimOTA generates about 10 times as many matches as O2O in extreme cases. This demonstrates that O2O has fewer positive matches, potentially slowing down optimization.

We propose **Dense O2O** as an efficient alternative. This strategy retains the one-to-one matching structure of O2O (with $M_i = 1$), but increases the number of targets (N) per image, achieving denser supervision. For example, as shown in Fig. 2c, we replicate the original image into four quadrants and combine them into a single composite image, maintaining the original image dimensions. This increases the number of targets from 1 to 4, boosting the supervision level in Eq. 1 while keeping the matching structure unchanged. Dense O2O achieves a level of supervision comparable to O2M but without the added complexity and computational overhead.

3.3. Improving matching quality: Matchability-Aware Loss

Limitations of VFL. The VariFocal Loss (VFL) [40], built on the FL [19], has been shown to improve object detection performance, especially in DETR models [2, 24, 43]. VFL loss is expressed as :

$$\text{VFL}(p, q, y) = \begin{cases} -q(q \log(p) + (1 - q) \log(1 - p)) & q > 0 \\ -\alpha p^\gamma \log(1 - p) & q = 0, \end{cases} \quad (3)$$

where q denotes the IoU between the predicted bounding box and its target box. For foreground samples ($q > 0$), the target label is set to q , while background samples ($q = 0$) have a target label of 0. VFL incorporates the IoU to improve the quality of queries in DETR [43].

However, VFL has two key limitations when optimizing low-quality matches: i). *Low-Quality Matches.* VFL focuses mainly on high-quality matches (high IoU). For low-quality matches (low IoU), the loss remains small, preventing the model from refining predictions for low-quality boxes. For low-quality matching (with low IoU, e.g., Fig. 2d), however, the loss remains minimal (marked by a $\star$ in Fig. 2e). ii) *Negative Samples.* VFL treats matches with no overlap as negative samples, which reduces the number of positive samples and limits effective training.

These issues are less problematic for traditional detectors due to their dense anchors and one-to-many assignment strategies. However, in the DETR framework, where queries are sparse and matching is more rigid, these limitations become more pronounced.

Matchability-Aware Loss. To address these issues, we propose the Matchability-Aware Loss (MAL), which extends the benefits of VFL while mitigating its shortcomings. MAL incorporates the matching quality directly into

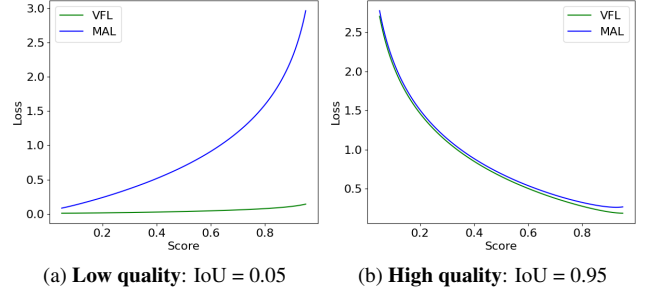


Figure 4. **VFL vs. MAL Comparison.** Comparison of VFL and our MAL for low-quality (IoU = 0.05, Fig. 4a) and high-quality (IoU = 0.95, Fig. 4b) matching cases.

the loss function, making it more sensitive to low-quality matches. The formula for MAL is:

$$\text{MAL}(p, q, y) = \begin{cases} -q^\gamma \log(p) - (1 - q^\gamma) \log(1 - p) & y = 1 \\ -p^\gamma \log(1 - p) & y = 0. \end{cases} \quad (4)$$

Compared to VFL, we introduce several small but important changes. Specifically, the target label has been modified from q to q^γ , simplifying the loss weights for positive and negative samples and removing the hyperparameter α used to balance positive and negative samples. This change helps to avoid the overemphasis on high-quality boxes and improves the overall training process. This can be easily seen from the loss landscape between VFL (in Fig. 2e) and MAL (in Fig. 2f). Note that the impact of γ is provided in Section 4.5.

Comparison with VFL. We compare MAL and VFL in handling both low-quality and high-quality matches. In the case of low-quality matches (IoU = 0.05, in Fig. 4a), MAL shows a sharper increase in loss as predicted confidence grows, compared to VFL, which remains almost unchanged. For high-quality matches (IoU = 0.95, in Fig. 4b), both MAL and VFL perform similarly, confirming that MAL improves training efficiency without compromising the performance on high-quality matches.

4. Experiments

4.1. Training details

For Dense O2O, we apply mosaic augmentation [1] and mixup augmentation [38] to generate additional positive samples per image. The impact of these augmentations is discussed in Section 4.5. We train our models on the MS-COCO dataset [20] using the AdamW optimizer [23]. Standard data augmentations, such as color jitter and zoom-out, are used, as in RT-DETR [24, 43] and D-FINE [27]. We employ a flat cosine learning rate scheduler [25] and propose a novel data augmentation scheduler. A data augmentation warmup strategy is used in the first few training epochs (four usually) for simplifying attention learning.

有时单张图像的正样本数量甚至超过80个。图3b进一步强调，在极端情况下，SimOTA生成的匹配数量约为O2O的10倍。这表明O2O的正匹配较少，可能会减缓优化进程。

我们提出密集一对一匹配（Dense O2O）作为一种高效的替代方案。该策略保留了一对一匹配的结构（其中 $M_i = 1$ ），但增加了每张图像中的目标数量（ N ），从而实现更密集的监督。例如，如图2c所示，我们将原始图像复制为四个象限，并将它们组合成一张复合图像，同时保持原始图像尺寸不变。这使得目标数量从1增加到4，从而提升了公式1中的监督强度，同时保持匹配结构不变。密集一对一匹配实现了与一对多匹配相当的监督水平，但避免了额外的复杂性和计算开销。

3.3. 提升匹配质量：可匹配性感知损失

VFL的局限性。基于FL [19]构建的Varifocal损失（VFL）[40]已被证明能提升目标检测性能，尤其在DETR模型[2, 24, 43]中。VFL损失表示为：

$$\text{VFL}(p, q, y) = \begin{cases} -q(q \log(p) + (1 - q) \log(1 - p)) & q > 0 \\ -\alpha p^\gamma \log(1 - p) & q = 0, \end{cases} \quad (3)$$

其中 q 表示预测边界框与其目标框之间的交并比。对于前景样本（ $q > 0$ ），目标标签设为 q ；而背景样本（ $q = 0$ ）的目标标签为 0。VFL 通过引入交并比来提升 DETR [43] 中查询的质量。

然而，VFL在优化低质量匹配时存在两个关键限制：i) *Low-Quality Matches*。VFL主要关注高质量匹配（高IoU）。对于低质量匹配（低IoU），损失值仍然很小，阻碍了模型对低质量框的预测进行优化。然而，对于低质量匹配（IoU较低，例如图2d），损失值仍然极小（在图2e中以*标记）。ii) *Negative Samples*。VFL将无重叠的匹配视为负样本，这减少了正样本的数量并限制了有效训练。

这些问题对于传统检测器来说不那么成问题，因为它们拥有密集的锚点和一对多的分配策略。然而，在DETR框架中，查询是稀疏的且匹配更为严格，这些局限性就变得更加明显。

匹配感知损失。为了解决这些问题，我们提出了匹配感知损失（MAL），它在延续VFL优势的同时，弥补了其不足。MAL将匹配质量直接纳入

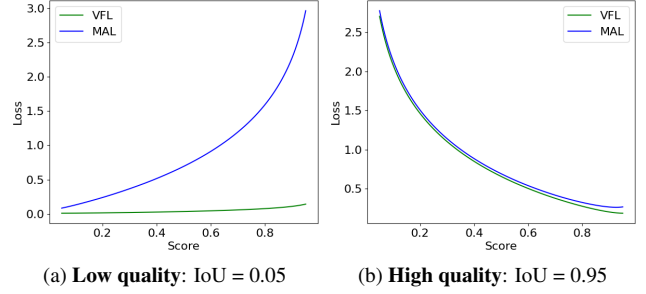


图4. VFL与MAL对比。针对低质量（IoU = 0.05，图4a）与高质量（IoU = 0.95，图4b）匹配案例，对比VFL与本文提出的MAL方法。

损失函数，使其对低质量匹配更加敏感。MAL的公式为：

$$\text{MAL}(p, q, y) = \begin{cases} -q^\gamma \log(p) - (1 - q^\gamma) \log(1 - p) & y = 1 \\ -p^\gamma \log(1 - p) & y = 0. \end{cases} \quad (4)$$

与VFL相比，我们引入了几处微小但重要的改动。具体而言，目标标签已从 q 修改为 q^γ ，简化了正负样本的损失权重，并移除了用于平衡正负样本的超参数 α 。这一改动有助于避免对高质量框的过度关注，并提升了整体训练效果。从VFL（图2e）与MAL（图2f）的损失曲面对比中可以清晰看出这一点。需注意， γ 的影响分析详见第4.5节。

与VFL的对比。我们比较了MAL和VFL在处理低质量匹配和高质量匹配时的表现。在低质量匹配的情况下（IoU = 0.05，见图4a），随着预测置信度的增加，MAL的损失值呈现更急剧的上升，而VFL的损失几乎保持不变。对于高质量匹配（IoU = 0.95，见图4b），MAL和VFL的表现相似，这证实了MAL在提升训练效率的同时，并未损害高质量匹配上的性能。

4. 实验

4.1. 训练细节

对于密集O2O，我们应用马赛克增强[1]和混合增强[38]来为每张图像生成额外的正样本。这些增强方法的影响在第4.5节中讨论。我们在MS-COCO数据集[20]上使用AdamW优化器[23]训练模型。如RT-DETR[24, 43]和D-FINE[27]中所述，采用了标准数据增强方法，如颜色抖动和缩小。我们采用平坦余弦学习率调度器[25]，并提出了一种新颖的数据增强调度器。在最初的几个训练周期（通常为四个）中，使用了数据增强预热策略，以简化注意力学习。

Table 1. **Comparison with real-time object detectors on COCO [20] val2017.** By integrating our method into D-FINE-L [27] and D-FINE-X [27], we build DEIM-D-FINE-L and DEIM-D-FINE-X. We compare our method with YOLO-based and DETR-based real-time object detectors. ★ indicates that the NMS is tuned with a confidence threshold of 0.01.

Model	#Epochs	#Params	GFLOPs	Latency (ms)	AP ^{val}	AP ^{val} ₅₀	AP ^{val} ₇₅	AP ^{val} _S	AP ^{val} _M	AP ^{val} _L
YOLO-based Real-time Object Detectors										
YOLOv8-L [12]	500	43	165	12.31	52.9	69.8	57.5	35.3	58.3	69.8
YOLOv8-X [12]	500	68	257	16.59	53.9	71.0	58.7	35.7	59.3	70.7
YOLOv9-C [34]	500	25	102	10.66	53.0	70.2	57.8	36.2	58.5	69.3
YOLOv9-E [34]	500	57	189	20.53	55.6	72.8	60.6	40.2	61.0	71.4
Gold-YOLO-L [33]	300	75	152	9.21	53.3	70.9	-	33.8	58.9	69.9
YOLOv10-L★ [32]	500	24	120	7.66	53.2	70.1	58.1	35.8	58.5	69.4
YOLOv10-X★ [32]	500	30	160	10.74	54.4	71.3	59.3	37.0	59.8	70.9
YOLO11-L★ [13]	500	25	87	6.31	52.9	69.4	57.7	35.2	58.7	68.8
YOLO11-X★ [13]	500	57	195	10.52	54.1	70.8	58.9	37.0	59.2	69.7
DETR-based Real-time Object Detectors										
RT-DETR-HG-L [43]	72	32	107	8.77	53.0	71.7	57.3	34.6	57.4	71.2
RT-DETR-HG-X [43]	72	67	234	13.51	54.8	73.1	59.4	35.7	59.6	72.9
D-FINE-L [27]	72	31	91	8.07	54.0	71.6	58.4	36.5	58.0	71.9
DEIM-D-FINE-L	50	31	91	8.07	54.7	72.4	59.4	36.9	59.6	71.8
D-FINE-X [27]	72	62	202	12.89	55.8	73.7	60.2	37.3	60.5	73.4
DEIM-D-FINE-X	50	62	202	12.89	56.5	74.0	61.5	38.8	61.4	74.2

Disabling Dense O2O after 50% of training epochs leads to better results. Following RT-DETRv2 [43], we turn off data augmentation in the last two epochs. Our LR and DataAug schedulers are depicted specifically in Fig. 5. Our backbones are pre-trained on ImageNet1k [8]. We evaluate our models on the MS-COCO validation set at a resolution of 640×640 . Additional details about the hyperparameters are provided in the supplementary material.

4.2. Comparisons with real-time detectors

We integrate our method into D-FINE-L [27] and D-FINE-X [27] building our DEIM-D-FINE-L and DEIM-D-FINE-X. We then evaluate these models and benchmark their real-time object detection performance against state-of-the-art models, including YOLOv8 [12], YOLOv9 [34], YOLOv10 [34], YOLOv11 [13], as well as DETR-based models like RT-DETRv2 [24] and D-FINE [27]. Tab. 1 compares the models in terms of epochs, parameters, GFLOPs, latency, and detection accuracy. Additional comparisons of smaller model variants (S and M) are included in the supplementary material.

Our method outperforms the current state-of-the-art models in training cost, inference latency, and detection accuracy, setting a new benchmark for real-time object detection. Note that D-FINE [27] is a very recent work that enhances the performance of RT-DETRv2 [24] by incorporating distillation and bounding box refinement, establishing itself as a leading real-time detector. Our DEIM further boosts the performance of D-FINE, achieving a 0.7 AP gain while reducing training costs by 30%, with no added inference latency. The most significant improvement is observed in small object detection, where D-FINE-X [27], when trained with our method, achieves a 1.5 AP gain as

DEIM-D-FINE-X.

When compared directly to YOLOv11-X [13], our DEIM-D-FINE-L outperforms this SoTA model, achieving slightly higher performance (54.7 vs. 54.1 AP) and reducing inference time by 20% (8.07 ms vs. 10.74 ms). Although YOLOv10 [34] uses a hybrid O2M and O2O assignment strategy, our models consistently outperform YOLOv10, demonstrating the effectiveness of Dense O2O.

Despite significant improvements in small object detection over other DETR-based models, our approach shows a slight decrease in small object AP compared to YOLO models. For example, YOLOv9-E [34] outperforms D-FINE-L [27] by approximately 1.4 AP on small objects, though our model achieves a higher overall AP (56.5 vs. 55.6). This gap underscores the ongoing challenges in small object detection within the DETR architecture and suggests potential areas for further improvement.

4.3. Comparisons with ResNet [14]-based DETRs

Most DETR research uses ResNet [14] as the backbone, and to enable a comprehensive comparison across existing DETR variants, we also applied our method to RT-DETRv2 [24], a state-of-the-art DETR variant. The results are summarized in Tab. 2. Unlike the original DETR, which requires 500 epochs for effective training, recent DETR variants, including ours, reduce training time while improving model performance. Our method shows the most significant improvements, surpassing all variants after just 36 epochs. Specifically, DEIM reduces training time by half and increases AP by 0.5 and 0.9 on RT-DETRv2 [24] with ResNet-50 [14] and ResNet-101 [14] backbones, respectively. Moreover, it outperforms DINO-Deformable-DETR [39] by 2.7 AP with the ResNet-50 [14] backbone.

表1. 在COCO [20] val2017数据集上与实时目标检测器的对比。通过将我们的方法集成到D-FINE-L [27]和D-FINE-X [27]中，我们构建了DEIM-D-FINE-L和DEIM-D-FINE-X。我们将我们的方法与基于YOLO和基于DETR的实时目标检测器进行了比较。★表示NMS使用0.01的置信度阈值进行调优。

Model	#Epochs	#Params	GFLOPs	Latency (ms)	AP ^{val}	AP ^{val} ₅₀	AP ^{val} ₇₅	AP ^{val} _S	AP ^{val} _M	AP ^{val} _L
YOLO-based Real-time Object Detectors										
YOLOv8-L [12]	500	43	165	12.31	52.9	69.8	57.5	35.3	58.3	69.8
YOLOv8-X [12]	500	68	257	16.59	53.9	71.0	58.7	35.7	59.3	70.7
YOLOv9-C [34]	500	25	102	10.66	53.0	70.2	57.8	36.2	58.5	69.3
YOLOv9-E [34]	500	57	189	20.53	55.6	72.8	60.6	40.2	61.0	71.4
Gold-YOLO-L [33]	300	75	152	9.21	53.3	70.9	-	33.8	58.9	69.9
YOLOv10-L* [32]	500	24	120	7.66	53.2	70.1	58.1	35.8	58.5	69.4
YOLOv10-X* [32]	500	30	160	10.74	54.4	71.3	59.3	37.0	59.8	70.9
YOLO11-L* [13]	500	25	87	6.31	52.9	69.4	57.7	35.2	58.7	68.8
YOLO11-X* [13]	500	57	195	10.52	54.1	70.8	58.9	37.0	59.2	69.7
DETR-based Real-time Object Detectors										
RT-DETR-HG-L [43]	72	32	107	8.77	53.0	71.7	57.3	34.6	57.4	71.2
RT-DETR-HG-X [43]	72	67	234	13.51	54.8	73.1	59.4	35.7	59.6	72.9
D-FINE-L [27]	72	31	91	8.07	54.0	71.6	58.4	36.5	58.0	71.9
DEIM-D-FINE-L	50	31	91	8.07	54.7	72.4	59.4	36.9	59.6	71.8
D-FINE-X [27]	72	62	202	12.89	55.8	73.7	60.2	37.3	60.5	73.4
DEIM-D-FINE-X	50	62	202	12.89	56.5	74.0	61.5	38.8	61.4	74.2

在训练周期达到50%后禁用密集O2O可获得更好的结果。遵循RT-DETRv2 [43]的做法，我们在最后两个周期关闭数据增强。我们的学习率与数据增强调度器具体如图5所示。我们的骨干网络均在ImageNet1k [8]上进行了预训练。我们在分辨率为640 × 640的MS-COCO验证集上评估模型。超参数的更多细节详见补充材料。

4.2. 与实时检测器的比较

我们将我们的方法集成到D-FINE-L [27]和D-FINE-X [27]中，构建了DEIM-D-FINE-L和DEIM-D-FINE-X。随后，我们评估了这些模型，并将其实时目标检测性能与先进模型进行基准测试，包括YOLOv8 [12]、YOLOv9 [34]、YOLOv10 [34]、YOLOv11 [13]，以及基于DETR的模型如RT-DETRv2 [24]和D-FINE [27]。表1从训练轮次、参数量、GFLOPs、延迟和检测精度等方面对这些模型进行了比较。补充材料中还包含了较小模型变体（S和M）的额外对比。

我们的方法在训练成本、推理延迟和检测精度上均超越了当前最先进的模型，为实时目标检测设立了新标杆。值得注意的是，D-FINE [27]作为一项最新研究，通过融合知识蒸馏与边界框优化技术，显著提升了RT-DETRv2 [24]的性能，已成为领先的实时检测器。我们的DEIM方法进一步强化了D-FINE的性能，在保持零额外推理延迟的同时，实现了0.7 AP的性能提升，并将训练成本降低30%。其中，小目标检测领域的提升最为显著：采用我们方法训练的D-FINE-X [27]获得了1.5 AP的性能增益。

DEIM-D-FINE-X。

与YOLOv11-X [13]直接相比，我们的DEIM-D-FINE-L超越了这一先进模型，实现了略高的性能（54.7 vs. 54.1 AP）并将推理时间降低了20%（8.07 ms vs. 10.74 ms）。尽管YOLOv10 [34]采用了混合O2M与O2O分配策略，我们的模型始终优于YOLOv10，这证明了密集O2O策略的有效性。

尽管在小型物体检测方面相比其他基于DETR的模型已有显著提升，但我们的方法在小型物体AP上仍较YOLO模型略有下降。例如，YOLOv9-E [34]在小型物体检测上比D-FINE-L [27]高出约1.4 AP，尽管我们的模型实现了更高的整体AP（56.5对比55.6）。这一差距凸显了DETR架构在小型物体检测领域持续面临的挑战，并指明了未来可改进的潜在方向。

4.3. 与基于ResNet [14]的DETR的比较

大多数DETR研究采用ResNet [14]作为主干网络，为了全面比较现有DETR变体，我们也将方法应用于当前最先进的DETR变体RT-DETRv2 [24]。结果汇总于表2。与原始DETR需要500个训练周期才能有效训练不同，包括我们方法在内的近期DETR变体在提升模型性能的同时显著缩短了训练时间。我们的方法仅用36个周期就展现出最显著的改进，超越了所有变体。具体而言，在采用ResNet-50 [14]和ResNet-101 [14]主干的RT-DETRv2 [24]上，DEIM将训练时间缩短一半，并分别将AP提升0.5和0.9。此外，在ResNet-50 [14]主干下，其性能以2.7 AP的优势超越了DINO-Deformable-DETR [39]。

Table 2. **Comparison with ResNet-based DETRs on COCO [20] val2017.** By integrating our method into ResNet50 [14] and ResNet101 [14], we build DEIM-RT-DETRv2-R50 and DEIM-RT-DETRv2-R101. We compare our method with competitive DETR-based object detectors that use ResNet50 [14] or ResNet101 [14] as backbones.

Model	#Epochs	#Params	GFLOPs	AP ^{val}	AP ₅₀ ^{val}	AP ₇₅ ^{val}	AP _S ^{val}	AP _M ^{val}	AP _L ^{val}
ResNet50 [14]-based									
DETR-DC5 [3]	500	41	187	43.3	63.1	45.9	22.5	47.3	61.1
Anchor-DETR-DC5 [35]	50	39	172	44.2	64.7	47.5	24.7	48.2	60.6
Conditional-DETR-DC5 [26]	108	44	195	45.1	65.4	48.5	25.3	49.0	62.2
Efficient-DETR [36]	36	35	210	45.1	63.1	49.1	28.3	48.4	59.0
SMCA-DETR [11]	108	40	152	45.6	65.5	49.1	25.9	49.3	62.6
Deformable-DETR [45]	50	40	173	46.2	65.2	50.0	28.8	49.2	61.7
DAB-Deformable-DETR [21]	50	48	195	46.9	66.0	50.8	30.1	50.4	62.5
DN-Deformable-DETR [18]	50	48	195	48.6	67.4	52.7	31.0	52.0	63.7
DINO-Deformable-DETR [39]	36	47	279	50.9	69.0	55.3	34.6	54.1	64.6
RT-DETR [43]	72	42	136	53.1	71.3	57.7	34.8	58.0	70.0
RT-DETRv2 [24]	72	42	136	53.4	71.6	57.4	36.1	57.9	70.8
DEIM-RT-DETRv2	36	42	136	53.9	71.7	58.6	36.7	58.9	70.9
DEIM-RT-DETRv2	60	42	136	54.3	72.3	58.8	37.5	58.7	70.8
ResNet101 [14]-based									
DETR-DC5 [3]	500	60	253	44.9	64.7	47.7	23.7	49.5	62.3
Anchor-DETR-DC5 [35]	50	-	-	45.1	65.7	48.8	25.8	49.4	61.6
Conditional-DETR-DC5 [26]	108	63	262	45.9	66.8	49.5	27.2	50.3	63.3
Efficient-DETR [36]	36	54	289	45.7	64.1	49.5	28.2	49.1	60.2
SMCA-DETR [11]	108	58	218	46.3	66.6	50.2	27.2	50.5	63.2
RT-DETR [43]	72	76	259	54.3	72.7	58.6	36.0	58.8	72.1
RT-DETRv2 [24]	72	76	259	54.3	72.8	58.8	35.8	58.8	72.1
DEIM-RT-DETRv2	36	76	259	55.2	73.3	59.9	37.8	59.6	72.8
DEIM-RT-DETRv2	60	76	259	55.5	73.5	60.3	37.9	59.9	73.0

Table 3. **Comparison of the D-FINE and when with our DEIM on CrowdHuman [30].** Both are trained with 120 epochs.

Method	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
D-FINE-L	56.0	87.2	59.4	29.0	46.1	54.6
w/ DEIM	57.5	87.6	62.9	33.2	48.7	55.7

DEIM also significantly enhances small-object detection. For example, while achieving comparable overall AP to RT-DETRv2 [24], our DEIM-RT-DETRv2-R50 surpasses RT-DETRv2 by 1.3 AP on small objects. This improvement is even more pronounced with the larger ResNet-101 backbone, where our DEIM-RT-DETRv2-R101 outperforms RT-DETRv2-R101 by 2.1 AP on small objects. Extending training to 72 epochs further improves overall performance, especially with the ResNet-50 backbone, indicating that smaller models benefit from additional training.

4.4. Comparisons on CrowdHuman

CrowdHuman [30] is a benchmark dataset designed to evaluate object detectors in dense crowd scenarios. We applied both D-FINE and our proposed method to the CrowdHuman dataset, following the configurations provided in the D-FINE. As shown in Tab. 3, our approach (D-FINE-L enhanced with DEIM) achieves a notable improvement of 1.5 AP over D-FINE-L. In particular, our method delivers a significant performance boost (greater than 3% improvement) on small objects (AP_s) and high-quality detections (AP₇₅), demonstrating its ability to detect objects more accurately in

Table 4. **Comparison of Dense O2O methods with different combinations of mosaic and mixup augmentation strategies.** The probability values denote the likelihood of applying mosaic and mixup in each mini-batch during training.

Mosaic Prob.	Mixup Prob.	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
Training 12 Epochs							
0.0	0.0	49.6	67.1	53.6	31.3	54.2	67.8
0.5	0.0	50.4	68.4	54.5	32.7	54.6	68.1
0.0	0.5	50.1	67.7	54.0	31.1	54.5	68.7
0.5	0.5	50.4	68.1	54.2	32.7	54.7	68.2
Training 24 Epochs							
0.0	0.0	51.7	69.5	55.8	32.8	56.4	69.7
0.5	0.0	51.9	70.1	55.9	34.9	56.1	69.3
0.0	0.5	51.5	69.4	55.5	33.2	56.3	69.3
0.5	0.5	52.5	70.6	56.7	34.9	57.1	70.1

challenging scenarios. Furthermore, this experiment underscores the strong generalization capability of DEIM across diverse datasets, confirming its robustness.

4.5. Analysis

In the following studies, we use RT-DETRv2 [24] paired with ResNet50 [14] to conduct experiments and report the performance on MS-COCO val2017 as the default setup unless otherwise specified.

Methods for achieving Dense O2O. We explored two approaches to implement Dense O2O: mosaic [1] and mixup [38]. Mosaic is a data augmentation that combines

表2. 在COCO [20] val2017数据集上与基于ResNet的DETR方法对比。通过将我们的方法集成到ResNet50 [14]和ResNet101 [14]中，我们构建了DEIM-RT-DETRv2-R50和DEIM-RT-DETRv2-R101。我们将我们的方法与使用ResNet50 [14]或ResNet101 [14]作为骨干网络的竞争性基于DETR的目标检测器进行了比较。

Model	#Epochs	#Params	GFLOPs	AP ^{val}	AP ^{val} ₅₀	AP ^{val} ₇₅	AP ^{val} _S	AP ^{val} _M	AP ^{val} _L
ResNet50 [14]-based									
DETR-DC5 [3]	500	41	187	43.3	63.1	45.9	22.5	47.3	61.1
Anchor-DETR-DC5 [35]	50	39	172	44.2	64.7	47.5	24.7	48.2	60.6
Conditional-DETR-DC5 [26]	108	44	195	45.1	65.4	48.5	25.3	49.0	62.2
Efficient-DETR [36]	36	35	210	45.1	63.1	49.1	28.3	48.4	59.0
SMCA-DETR [11]	108	40	152	45.6	65.5	49.1	25.9	49.3	62.6
Deformable-DETR [45]	50	40	173	46.2	65.2	50.0	28.8	49.2	61.7
DAB-Deformable-DETR [21]	50	48	195	46.9	66.0	50.8	30.1	50.4	62.5
DN-Deformable-DETR [18]	50	48	195	48.6	67.4	52.7	31.0	52.0	63.7
DINO-Deformable-DETR [39]	36	47	279	50.9	69.0	55.3	34.6	54.1	64.6
RT-DETR [43]	72	42	136	53.1	71.3	57.7	34.8	58.0	70.0
RT-DETRv2 [24]	72	42	136	53.4	71.6	57.4	36.1	57.9	70.8
DEIM-RT-DETRv2	36	42	136	53.9	71.7	58.6	36.7	58.9	70.9
DEIM-RT-DETRv2	60	42	136	54.3	72.3	58.8	37.5	58.7	70.8
ResNet101 [14]-based									
DETR-DC5 [3]	500	60	253	44.9	64.7	47.7	23.7	49.5	62.3
Anchor-DETR-DC5 [35]	50	-	-	45.1	65.7	48.8	25.8	49.4	61.6
Conditional-DETR-DC5 [26]	108	63	262	45.9	66.8	49.5	27.2	50.3	63.3
Efficient-DETR [36]	36	54	289	45.7	64.1	49.5	28.2	49.1	60.2
SMCA-DETR [11]	108	58	218	46.3	66.6	50.2	27.2	50.5	63.2
RT-DETR [43]	72	76	259	54.3	72.7	58.6	36.0	58.8	72.1
RT-DETRv2 [24]	72	76	259	54.3	72.8	58.8	35.8	58.8	72.1
DEIM-RT-DETRv2	36	76	259	55.2	73.3	59.9	37.8	59.6	72.8
DEIM-RT-DETRv2	60	76	259	55.5	73.5	60.3	37.9	59.9	73.0

表3. 在CrowdHuman数据集[30]上，D-FINE与结合我们DEIM方法的性能对比。两者均训练120个周期。

Method	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
D-FINE-L	56.0	87.2	59.4	29.0	46.1	54.6
w/ DEIM	57.5	87.6	62.9	33.2	48.7	55.7

DEIM还显著提升了小目标检测性能。例如，在整体AP指标与RT-DETRv2 [24]相当的情况下，我们的DEIM-RT-DETRv2-R50模型在小目标检测上比RT-DETRv2高出1.3 AP。这一优势在使用更大的ResNet-101骨干网络时更为明显：DEIM-RT-DETRv2-R101在小目标检测上以2.1 AP的优势超越RT-DETRv2-R101。将训练周期延长至72轮可进一步提升整体性能，尤其在ResNet-50骨干网络上效果显著，这表明较小模型能从额外训练中获得更大收益。

4.4. CrowdHuman数据集上的比较

CrowdHuman [30]是一个为评估密集人群场景中目标检测器而设计的基准数据集。我们按照D-FINE提供的配置，将D-FINE和我们提出的方法应用于CrowdHuman数据集。如表3所示，我们的方法（通过DEIM增强的D-FINE-L）相比D-FINE-L实现了1.5 AP的显著提升。特别地，我们的方法在小目标检测（AP_s）和高精度检测（AP₇₅）上带来了明显的性能提升（改进幅度超过3%），这证明了其在密集场景中更准确检测目标的能力。

表4. 不同马赛克与混合增强策略组合的密集O2O方法比较。概率值表示训练期间每个小批量应用马赛克与混合增强的可能性。

Mosaic Prob.	Mixup Prob.	AP	AP ₅₀	AP ₇₅	AP _s	AP _m	AP _l
Training 12 Epochs							
0.0	0.0	49.6	67.1	53.6	31.3	54.2	67.8
0.5	0.0	50.4	68.4	54.5	32.7	54.6	68.1
0.0	0.5	50.1	67.7	54.0	31.1	54.5	68.7
0.5	0.5	50.4	68.1	54.2	32.7	54.7	68.2
Training 24 Epochs							
0.0	0.0	51.7	69.5	55.8	32.8	56.4	69.7
0.5	0.0	51.9	70.1	55.9	34.9	56.1	69.3
0.0	0.5	51.5	69.4	55.5	33.2	56.3	69.3
0.5	0.5	52.5	70.6	56.7	34.9	57.1	70.1

具有挑战性的场景。此外，该实验强调了DEIM在不同数据集间的强大泛化能力，证实了其稳健性。

4.5. 分析

在以下研究中，我们采用RT-DETRv2 [24] 与ResNet50 [14] 的组合进行实验，并默认在MS-COCO val2017数据集上报告性能，除非另有说明。

实现密集O2O的方法。我们探索了两种实现密集O2O的途径：马赛克[1]和混合[38]。马赛克是一种数据增强技术，通过组合

Table 5. **Impact of γ in MAL(Eqn. 4).** We report the performance on COCO [20] val2017 for 24 epochs.

γ	1.3	1.5	1.8	2.0
AP	52.2	52.4	52.1	51.9

Table 6. **Impact of Dense O2O and MAL.** We conduct experiments with RT-DETRv2-R50 [24] and D-FINE-L [27].

Epochs	Dense O2O	MAL	AP	AP ₅₀	AP ₇₅
RT-DETRv2-R50 [24]					
72			53.4	71.6	57.4
36	✓		53.6	71.9	58.2
	✓	✓	53.9	71.7	58.6
D-FINE-L [27]					
72			54.0	71.6	58.4
36	✓		54.2	72.1	58.9
	✓	✓	54.6	72.2	59.5

four images into one, while mixup overlays two images at a random ratio. Both methods effectively increase the number of targets per image, enhancing supervision during training.

As shown in Tab. 4, both mosaic and mixup lead to significant improvements after 12 epochs compared to training without target augmentation, highlighting the effectiveness of Dense O2O. Moreover, combining mosaic and mixup accelerates model convergence, further emphasizing the benefits of augmented supervision. We further tracked the number of positive samples per image over one training epoch, with results shown in Fig. 6. Compared to traditional O2O, Dense O2O significantly increases positive samples.

Overall, Dense O2O intensifies supervision by increasing target counts per image, leading to faster model convergence. Mosaic and mixup are simple, computationally efficient techniques that achieve this goal, and their effectiveness suggests further potential for exploring other methods to augment target counts during training.

Impact of γ in MAL(Eqn. 4). The results in Table 5 show the effect of different γ values on MAL after 24 epochs. Based on these experiments, we empirically set γ to 1.5, as it yields the best performance.

Effectiveness of Dense O2O and MAL. Tab. 6 presents the effectiveness of the two core components: Dense O2O and MAL. Dense O2O significantly accelerates model convergence, achieving performance similar to the baseline after just 36 epochs, as opposed to the 72 epochs required for the original model. When combined with MAL, our method further improves performance. This improvement is largely driven by better box quality, aligning with our goal of optimizing low-quality matches to improve high-quality box predictions. Overall, Dense O2O and MAL consistently deliver performance gains across both RT-DETRv2 and D-FINE, demonstrating their robustness and generalizability.

Table 7. **Training time in GPU hours.**

Method	Epoch	#GPU hr	AP
RT-DETRv2-R50	1	1.181	-
w/ DEIM	1	1.183	-
RT-DETRv2-R50	72	~85	53.4
w/ DEIM	60	~71	54.3

Table 8. **Fine-tuned results from Object365 pre-training.**

Method	Epoch	AP	AP ₅₀	AP ₇₅
D-FINE-X	32	59.3	76.8	64.6
w/ DEIM	24	59.5	76.4	65.2

Training speed. We provide an efficient implementation using Mosaic with caching and Mixup within batches. Tab. 7 shows the one-epoch training time on a single 4090 GPU, where DEIM is almost as fast as the baseline (1.183 vs. 1.181) and requires less training time to converge (71 vs. 85 hours). This highlights that our approach improves convergence while maintaining efficiency.

Finetuning from Object365. We directly employ the pre-trained Object365 weights from D-FINE and compare the results of fine-tuning with and without DEIM. As shown in Tab. 8, DEIM achieves better performance with fewer fine-tuning epochs. It further validates that DEIM delivers consistent gains, even when pre-trained on larger datasets.

5. Conclusion

In this paper, we present DEIM, a method designed to accelerate convergence in DETR-based real-time object detectors by improving matching. DEIM integrates Dense O2O matching, which increases the number of positive samples per image, with MAL, a novel loss designed to optimize matches across varying quality and specifically enhance low-quality matches. This combination substantially improves training efficiency, allowing DEIM to achieve superior performance in fewer epochs compared to models such as YOLOv11. DEIM demonstrates clear advantages over SoTA DETR models like RT-DETR and D-FINE, showing measurable gains in detection accuracy and training speed without compromising inference latency. These attributes establish DEIM as a highly effective solution for real-time applications, with the potential for further refinement and application across other high-performance detection tasks.

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表5. γ 在MAL（公式4）中的影响。我们在COCO [20] val2017上报告了24个epoch的性能。

γ	1.3	1.5	1.8	2.0
AP	52.2	52.4	52.1	51.9

表6. 密集O2O和MAL的影响。我们使用RT-DETRv2-R50 [24]和D-FINE-L [27]进行了实验。

Epochs	Dense O2O	MAL	AP	AP ₅₀	AP ₇₅
RT-DETRv2-R50 [24]					
72			53.4	71.6	57.4
36	✓		53.6	71.9	58.2
	✓	✓	53.9	71.7	58.6
D-FINE-L [27]					
72			54.0	71.6	58.4
36	✓		54.2	72.1	58.9
	✓	✓	54.6	72.2	59.5

四张图像合为一张，而mixup则以随机比例叠加两张图像。两种方法都有效增加了每张图像的目标数量，从而增强了训练过程中的监督效果。

如表4所示，与未使用目标增强的训练相比，经过12个周期后，马赛克和混合增强均带来显著提升，凸显了密集O2O的有效性。此外，结合马赛克与混合增强能加速模型收敛，进一步强调了增强监督的优势。我们进一步统计了单个训练周期内每张图像的正样本数量，结果如图6所示。相比传统O2O方法，密集O2O显著增加了正样本数量。

总体而言，密集O2O通过增加每张图像中的目标数量来加强监督，从而加快模型收敛速度。Mosaic和mixup是简单且计算效率高的技术，能够实现这一目标，它们的有效性表明，在训练期间探索其他方法来增加目标数量具有进一步潜力。

γ 在MAL中的影响（公式4）。表5中的结果展示了不同 γ 值在24个训练周期后对MAL的影响。基于这些实验，我们经验性地将 γ 设为1.5，因为它能带来最佳性能。

Dense O2O与MAL的有效性。表6展示了两个核心组件Dense O2O与MAL的有效性。Dense O2O显著加速了模型收敛，仅需36个周期即可达到与基线相当的性能，而原始模型需要72个周期。当与MAL结合时，我们的方法进一步提升了性能。这一改进主要得益于更优的边界框质量，这与我们通过优化低质量匹配来提升高质量边界框预测的目标一致。总体而言，Dense O2O和MAL在RT-DETRv2和D-FINE模型上均能持续带来性能提升，证明了其鲁棒性与泛化能力。

表7. GPU小时训练时间。

Method	Epoch	#GPU hr	AP
RT-DETRv2-R50	1	1.181	-
w/ DEIM	1	1.183	-
RT-DETRv2-R50	72	~85	53.4
w/ DEIM	60	~71	54.3

表8. 基于Object365预训练的微调结果。

Method	Epoch	AP	AP ₅₀	AP ₇₅
D-FINE-X	32	59.3	76.8	64.6
w/ DEIM	24	59.5	76.4	65.2

训练速度。我们提供了一个使用Mosaic的高效实现，支持批内缓存和Mixup。表7展示了在单块4090 GPU上的一轮训练时间，其中DEIM几乎与基线一样快（1.183对比1.181），且收敛所需训练时间更短（71小时对比85小时）。这凸显了我们的方法在保持效率的同时提升了收敛性。

基于Object365的微调。我们直接采用D-FINE中预训练的Object365权重，并比较使用与不使用DEIM的微调结果。如表8所示，DEIM以更少的微调轮次实现了更优性能。这进一步验证了即使在大规模数据集上预训练，DEIM仍能带来稳定的性能提升。

5. 结论

本文提出DEIM方法，旨在通过改进匹配机制加速基于DETR的实时目标检测器的收敛速度。DEIM融合了密集一对一匹配（增加每张图像的正样本数量）与新型损失函数MAL（专门针对不同质量匹配进行优化，尤其强化低质量匹配）。这种组合显著提升了训练效率，使DEIM相比YOLOv11等模型能以更少训练轮次达到更优性能。相较于RT-DETR、D-FINE等前沿DETR模型，DEIM在检测精度与训练速度上均展现出显著优势，且未增加推理延迟。这些特性使DEIM成为实时应用场景的高效解决方案，并具备进一步优化及拓展至其他高性能检测任务的潜力。

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DEIM: DETR with Improved Matching for Fast Convergence

Supplementary Material

1. Experimental Settings

Dataset and metric. We evaluate our method on the COCO [20] dataset, training DEIM on `train2017` and validating it on `val2017`. Standard COCO metrics are reported, including AP (averaged over IoU thresholds from 0.50 to 0.95 with a step size of 0.05), AP_{50} , AP_{75} , and AP at different object scales: AP_S , AP_M , and AP_L .

Table 9. Different hyperparameters for D-FINE models trained with DEIM.

D-FINE	X	L	M	S
Base LR	5e-4	5e-4	4e-4	4e-4
Min LR	2.5e-4	2.5e-4	2e-4	2e-4
Backbone LR	5e-6	2.5e-5	4e-5	2e-4
Backbone MinLR	2.5e-6	1.25e-5	2e-5	1e-4
Weight of MAL	1	1	1	1
γ in MAL	1.5	1.5	1.5	1.5
Freeze Backbone BN	False	False	False	False
Decoder Act.	SiLU	SiLU	SiLU	SiLU
Epochs	50	50	90	120

Table 10. Different hyperparameters for RT-DETRv2 models trained with DEIM.

RT-DETRv2	X	L	M*	M	S
Base LR	2e-4	2e-4	2e-4	2e-4	2e-4
Min LR	1e-4	1e-4	1e-4	1e-4	1e-4
Backbone LR	2e-6	2e-5	2e-5	1e-4	2e-4
Backbone MinLR	1e-6	1e-5	1e-5	5e-5	1e-4
Weight of MAL	1	1	1	1	1
γ in MAL	1.5	1.5	1.5	1.5	1.5
Freeze Backbone BN	False	False	False	False	False
Decoder Act.	SiLU	SiLU	SiLU	SiLU	SiLU
Epochs	60	60	60	120	120

Implementation details. We implement and validate our method using the D-FINE [27] and RT-DETRv2 [24, 43] frameworks. Most hyperparameters follow their original settings, with differences detailed in Tab. 9 and Tab. 10, respectively. Inspired by the FlatCosine LR scheduler in RTMDet [25], we propose a novel data augmentation scheduler tailored for Dense O2O. Attention mechanisms in DETRs are critical for extracting accurate object features for localization and classification. However, learning attention from scratch without inductive biases can be challenging. To mitigate this, we introduce a data augmentation warmup strategy, referred to as DataAug Warmup, which simplifies

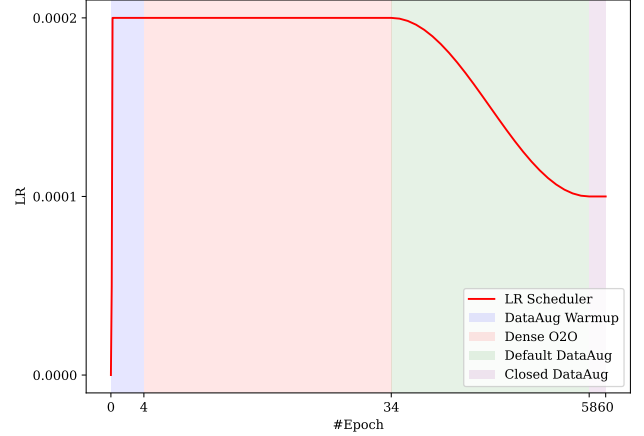


Figure 5. An illustrated example of our proposed novel training scheme for learning rate and data augmentation scheduler.

the learning by disabling advanced data augmentations during the initial epochs. An example of the FlatCosine LR and proposed DataAug schedulers for 60 training epochs is shown in Fig. 5.

2. Comparison with Lighter YOLO Detectors

We present the results of comparisons with more lightweight real-time models (S and M sizes) in the Table 11. Based on the strong real-time detectors RT-DETRv2 [24] and D-FINE [27], our DEIM achieves significant improvements across the board. Notably, in RT-DETRv2, all three model sizes show an approximately 1 AP improvement, with the DEIM-RT-DETRv2-M* achieving a remarkable 1.3 AP gain. Compared to other methods, our approach achieves the latest state-of-the-art results.

3. Additional Results

Effectiveness of the minor modifications. We incorporate minor modifications, including unfreezing the BN layers in the Backbone, adopting the FlatCosine LR scheduler, and replacing the Decoder activation function with SiLU, into both D-FINE-L and D-FINE-X. After training for 36 epochs, we observe that these changes have no impact on D-FINE-L but lead to a 0.1 AP improvement for D-FINE-X (55.4 vs. 55.5). This configuration is used as the new baseline for our experiments.

Number of positive samples between with/without Dense O2O. During one epoch of training, we compared the number of positive samples in the same training images

DEIM: 具有改进匹配机制的DETR, 实现快速收敛

补充材料

1. 实验设置

数据集与评估指标。我们在COCO[20]数据集上评估所提方法, 使用train2017训练DEIM并在val2017上进行验证。报告采用标准COCO评估指标, 包括AP (在IoU阈值0.50至0.95范围内以0.05为步长的平均值)、AP₅₀、AP₇₅, 以及不同物体尺度下的AP: AP_S、AP_M和AP_L。

表9. 使用DEIM训练的D-FINE模型的不同超参数。

D-FINE	X	L	M	S
Base LR	5e-4	5e-4	4e-4	4e-4
Min LR	2.5e-4	2.5e-4	2e-4	2e-4
Backbone LR	5e-6	2.5e-5	4e-5	2e-4
Backbone MinLR	2.5e-6	1.25e-5	2e-5	1e-4
Weight of MAL	1	1	1	1
γ in MAL	1.5	1.5	1.5	1.5
Freeze Backbone BN	False	False	False	False
Decoder Act.	SiLU	SiLU	SiLU	SiLU
Epochs	50	50	90	120

表10. 使用DEIM训练的RT-DETRv2模型的不同超参数。

RT-DETRv2	X	L	M*	M	S
Base LR	2e-4	2e-4	2e-4	2e-4	2e-4
Min LR	1e-4	1e-4	1e-4	1e-4	1e-4
Backbone LR	2e-6	2e-5	2e-5	1e-4	2e-4
Backbone MinLR	1e-6	1e-5	1e-5	5e-5	1e-4
Weight of MAL	1	1	1	1	1
γ in MAL	1.5	1.5	1.5	1.5	1.5
Freeze Backbone BN	False	False	False	False	False
Decoder Act.	SiLU	SiLU	SiLU	SiLU	SiLU
Epochs	60	60	60	120	120

实现细节。我们使用D-FINE [27]和RT-DETRv2 [24, 43]框架实现并验证了我们的方法。大多数超参数遵循其原始设置, 具体差异详见表9和表10。受RT-MDet [25]中FlatCosine学习率调度器的启发, 我们提出了一种专为密集O2O设计的新型数据增强调度器。DETR中的注意力机制对于提取准确的目标定位和分类特征至关重要。然而, 在没有归纳偏置的情况下从头学习注意力可能具有挑战性。为了缓解这一问题, 我们引入了一种数据增强预热策略, 称为DataAug Warmup, 它简化了

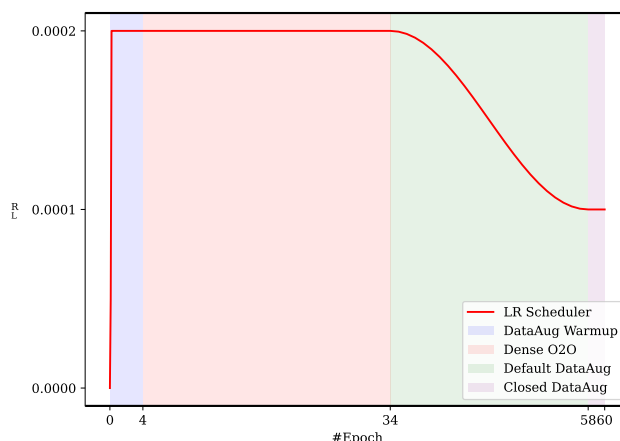


图5. 我们提出的用于学习率和数据增强调度器的新型训练方案示意图。

通过在初始训练周期禁用高级数据增强来实现学习。图5展示了针对60个训练周期的FlatCosine学习率调度器及所提出的数据增强调度器的示例。

2. 与轻量化YOLO检测器的比较

我们在表11中展示了与更轻量级实时模型 (S和M尺寸) 的对比结果。基于强大的实时检测器RT-DETRv2 [24]和D-FINE [27], 我们的DEIM模型实现了全方位的显著提升。值得注意的是, 在RT-DETRv2中, 所有三种模型尺寸均显示出约1 AP的改进, 其中DEIM-RT-DETRv2-M*取得了1.3 AP的显著增益。与其他方法相比, 我们的方法达到了最新的最先进水平。

3. 补充结果

微小修改的有效性。我们将包括解冻主干网络中的BN层、采用FlatCosine学习率调度器以及将解码器激活函数替换为SiLU在内的微小修改, 同时应用于D-FINE-L和D-FINE-X。经过36个周期的训练后, 我们观察到这些改动对D-FINE-L没有影响, 但使D-FINE-X的AP提升了0.1 (55.4对比55.5)。此配置将作为我们实验的新基准。

带/不带密集一对一 (Dense O2O) 时的正样本数量对比。在单轮训练周期中, 我们比较了相同训练图像中正样本的数量。

Table 11. **Comparison with S and M sized real-time object detectors on COCO [20] val_{2017} .** $\star$ indicates that the NMS is tuned with a confidence threshold of 0.01.

Model	#Epochs	#Params.	GFLOPs	Latency (ms)	AP^{val}	AP_{50}^{val}	AP_{75}^{val}	AP_S^{val}	AP_M^{val}	AP_L^{val}
YOLO-based Real-time Object Detectors										
YOLOv8-S [12]	500	11	29	6.96	44.9	61.8	48.6	25.7	49.9	61.0
YOLOv8-M [12]	500	26	79	9.66	50.2	67.2	54.6	32.0	55.7	66.4
YOLOv9-S [34]	500	7	26	8.02	46.8	61.8	48.6	25.7	49.9	61.0
YOLOv9-M [34]	500	20	76	10.15	51.4	67.2	54.6	32.0	55.7	66.4
Gold-YOLO-S [33]	300	22	46	2.01	46.4	63.4	-	25.3	51.3	63.6
Gold-YOLO-M [33]	300	41	88	3.21	51.1	68.5	-	32.3	56.1	68.6
YOLOv10-S [32]	500	7	22	2.65	46.3	63.0	50.4	26.8	51.0	63.8
YOLOv10-M [32]	500	15	59	4.97	51.1	68.1	55.8	33.8	56.5	67.0
YOLO11-S $\star$ [13]	500	9	22	2.86	47.0	63.9	50.7	29.0	51.7	64.4
YOLO11-M $\star$ [13]	500	20	68	4.95	51.5	68.5	55.7	33.4	57.1	67.9
DETR-based Real-time Object Detectors										
RT-DETR-R18 [43]	72	20	61	4.63	46.5	63.8	50.4	28.4	49.8	63.0
RT-DETR-R34 [43]	72	31	93	6.43	48.9	66.8	52.9	30.6	52.4	66.3
RT-DETRv2-S [24]	120	20	60	4.59	48.1	65.1	57.4	36.1	57.9	70.8
DEIM-RT-DETRv2-S	120	20	60	4.59	49.0	66.1	53.3	32.6	52.5	64.1
RT-DETRv2-M [24]	120	31	92	6.40	49.9	67.5	58.6	35.8	58.6	72.1
DEIM-RT-DETRv2-M	120	31	92	6.40	50.9	68.6	55.2	34.3	54.4	67.1
RT-DETRv2-M $\star$ [24]	72	33	100	6.90	51.9	69.9	56.5	33.5	56.8	69.2
DEIM-RT-DETRv2-M$\star$	60	33	100	6.90	53.2	71.2	57.8	35.3	57.6	70.2
D-FINE-Nano [27]	148	4	7	2.12	42.8	60.3	45.5	22.9	46.8	62.1
DEIM-D-FINE-Nano	148	4	7	2.12	43.0	60.4	46.2	24.5	47.1	62.1
D-FINE-S [27]	120	10	25	3.49	48.5	65.6	52.6	29.1	52.2	65.4
DEIM-D-FINE-S	120	10	25	3.49	49.0	65.9	53.1	30.4	52.6	65.7
D-FINE-M [27]	120	19	57	5.55	52.3	69.8	56.4	33.2	56.5	70.2
DEIM-D-FINE-M	90	19	57	5.55	52.7	70.0	57.3	35.3	56.7	69.5

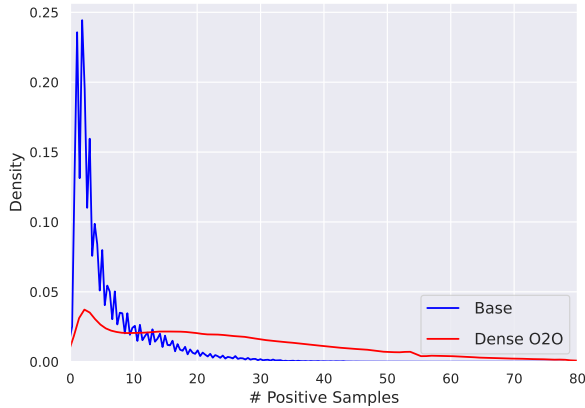


Figure 6. **# Positive Samples with and without Dense O2O in One Epoch of Training.** *Base* indicates without Dense O2O.

with and without using Dense O2O, as shown in Fig. 6. After incorporating Dense O2O, the number of positive samples significantly increases. This further supports our claim that Dense O2O effectively enhances supervision.

Studies of the number of positive samples. We adjust the average number of objects per image during training by

Table 12. **Varying the number of objects per training image.**

Avg # objects	AP	AP_{50}	AP_{75}
Training 24 Epochs			
~ 10	51.7	69.5	55.8
~ 25	52.5	70.6	56.7
~ 50	52.2	70.1	56.4

Table 13. **Training and validation accuracy.**

Model	AP_{train}	AP_{val}
RT-DETRv2-R50	65.1	53.4
w/ DEIM	64.8	54.3

modifying Dense O2O. As shown in Tab. 12, performance improves significantly when the number increases from 10 (without Dense O2O) to 25 (Default Dense O2O) but drops at 50 (Max Dense O2O). This decline is likely due to an imbalance in the positive-to-negative ratio and a data distribution shift caused by too many objects. Notably, an average of 25 objects aligns with the default experimental setting

表 11. 在 COCO [20] val2017 数据集上与 S 和 M 尺寸实时目标检测器的比较。* 表示 NMS 经过调优
置置信度阈值为0.01。th

Model	#Epochs	#Params.	GFLOPs	Latency (ms)	AP^{val}	AP_{50}^{val}	AP_{75}^{val}	AP_S^{val}	AP_M^{val}	AP_L^{val}
YOLO-based Real-time Object Detectors										
YOLOv8-S [12]	500	11	29	6.96	44.9	61.8	48.6	25.7	49.9	61.0
YOLOv8-M [12]	500	26	79	9.66	50.2	67.2	54.6	32.0	55.7	66.4
YOLOv9-S [34]	500	7	26	8.02	46.8	61.8	48.6	25.7	49.9	61.0
YOLOv9-M [34]	500	20	76	10.15	51.4	67.2	54.6	32.0	55.7	66.4
Gold-YOLO-S [33]	300	22	46	2.01	46.4	63.4	-	25.3	51.3	63.6
Gold-YOLO-M [33]	300	41	88	3.21	51.1	68.5	-	32.3	56.1	68.6
YOLOv10-S [32]	500	7	22	2.65	46.3	63.0	50.4	26.8	51.0	63.8
YOLOv10-M [32]	500	15	59	4.97	51.1	68.1	55.8	33.8	56.5	67.0
YOLO11-S* [13]	500	9	22	2.86	47.0	63.9	50.7	29.0	51.7	64.4
YOLO11-M* [13]	500	20	68	4.95	51.5	68.5	55.7	33.4	57.1	67.9
DETR-based Real-time Object Detectors										
RT-DETR-R18 [43]	72	20	61	4.63	46.5	63.8	50.4	28.4	49.8	63.0
RT-DETR-R34 [43]	72	31	93	6.43	48.9	66.8	52.9	30.6	52.4	66.3
RT-DETRv2-S [24]	120	20	60	4.59	48.1	65.1	57.4	36.1	57.9	70.8
DEIM-RT-DETRv2-S	120	20	60	4.59	49.0	66.1	53.3	32.6	52.5	64.1
RT-DETRv2-M [24]	120	31	92	6.40	49.9	67.5	58.6	35.8	58.6	72.1
DEIM-RT-DETRv2-M	120	31	92	6.40	50.9	68.6	55.2	34.3	54.4	67.1
RT-DETRv2-M* [24]	72	33	100	6.90	51.9	69.9	56.5	33.5	56.8	69.2
DEIM-RT-DETRv2-M*	60	33	100	6.90	53.2	71.2	57.8	35.3	57.6	70.2
D-FINE-Nano [27]	148	4	7	2.12	42.8	60.3	45.5	22.9	46.8	62.1
DEIM-D-FINE-Nano	148	4	7	2.12	43.0	60.4	46.2	24.5	47.1	62.1
D-FINE-S [27]	120	10	25	3.49	48.5	65.6	52.6	29.1	52.2	65.4
DEIM-D-FINE-S	120	10	25	3.49	49.0	65.9	53.1	30.4	52.6	65.7
D-FINE-M [27]	120	19	57	5.55	52.3	69.8	56.4	33.2	56.5	70.2
DEIM-D-FINE-M	90	19	57	5.55	52.7	70.0	57.3	35.3	56.7	69.5

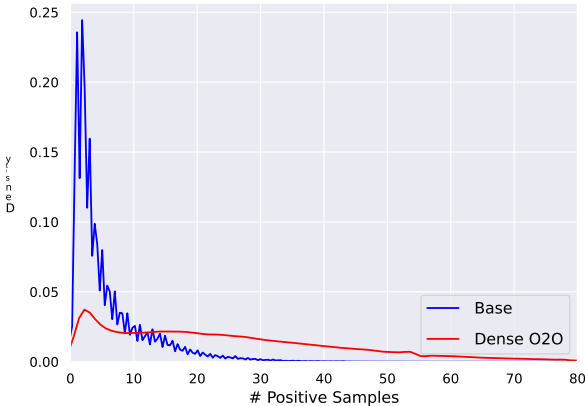


图6. 训练单周期内使用与不使用密集O2O的正样本数量对比。*Base*表示未使用密集O2O。

在使用和不使用Dense O2O的情况下，如图6所示。在引入Dense O2O后，正样本数量显著增加。这进一步支持了我们的主张，即Dense O2O能有效增强监督。

正样本数量的研究。我们通过在训练期间调整每张图像的平均物体数量来实现

表12. 调整每张训练图像中的物体数量。

Avg # objects	AP	AP_{50}	AP_{75}
Training 24 Epochs			
~ 10	51.7	69.5	55.8
~ 25	52.5	70.6	56.7
~ 50	52.2	70.1	56.4

表13. 训练和验证准确率。

Model	AP_{train}	AP_{val}
RT-DETRv2-R50	65.1	53.4
w/ DEIM	64.8	54.3

修改密集O2O。如表12所示，当数量从10（无密集O2O）增加到25（默认密集O2O）时，性能显著提升，但在50（最大密集O2O）时下降。这种下降可能是由于正负样本比例失衡以及过多物体导致的数据分布偏移所致。值得注意的是，平均25个物体与默认实验设置相符。

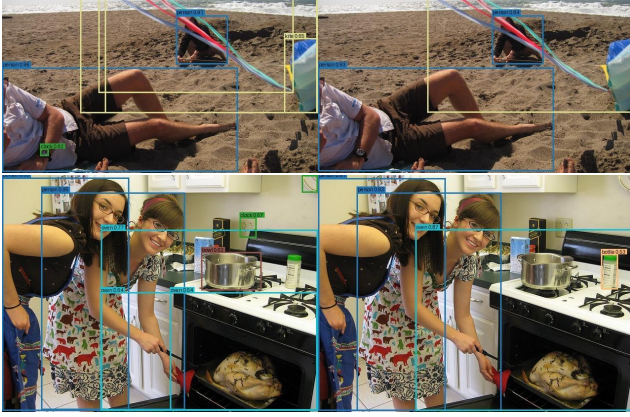


Figure 7. **Qualitative Comparison between D-FINE-L and DEIM.** In each paired image, the left is from D-FINE-L while the right is predicted by DEIM-D-FINE-L (Score threshold = 0.5).

used in this study, corresponding to the default Dense O2O configuration.

Training vs. validation accuracy. As shown in Tab. 13, DEIM achieves higher validation accuracy and slightly lower training accuracy, indicating reduced overfitting on the training set and improved adaptability to new samples.

4. Visualizations

We present the qualitative comparison results in Fig. 7. These results demonstrate that DEIM effectively addresses two critical issues faced by D-FINE-L: high-confidence duplicated predictions and false positives. For example, in the top row, a single kite is erroneously assigned four highly overlapping bounding boxes, each with high confidence scores. Furthermore, as shown in the bottom row, D-FINE-L misclassifies a socket and a wall-mounted object as a clock while failing to detect the bottle. By incorporating DEIM during training, the detector successfully resolves these challenges. This visualization highlights the significant advancements enabled by DEIM, underscoring its potential for improving detection accuracy.



图7. D-FINE-L与DEIM的定性比较。每组配对图像中，左侧为D-FINE-L结果，右侧为DEIM-D-FINE-L预测结果（得分阈值= 0.5）。

本研究中使用，对应于默认的Dense O2O配置。

训练与验证准确率。如表13所示，DEIM获得了更高的验证准确率和略低的训练准确率，这表明其在训练集上的过拟合减少，对新样本的适应性提高。

4. 可视化

我们在图7中展示了定性比较结果。这些结果表明，DEIM有效解决了D-FINE-L面临的两个关键问题：高置信度的重复预测和误报。例如，在第一行中，单个风筝被错误地分配了四个高度重叠的边界框，每个框都具有高置信度分数。此外，如最后一行所示，D-FINE-L将插座和壁挂物体误分类为时钟，同时未能检测到瓶子。通过在训练中引入DEIM，检测器成功解决了这些挑战。这一可视化结果突显了DEIM带来的显著进步，彰显了其提升检测准确性的潜力。